Software Requirements Specification

For

TripEasy Tourism Management System

Day Visit Planner for Gampaha

Version 1.1 approved

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Revision History

Name	Date	Reason for Changes	Version
E. L. N. Perera	12-Jul-2026	Initial draft based on approved project proposal	1.0
E. L. N. Perera	06-Sep-2026	Comprehensive revision based on feedback (Aspects 1–6): Added Section 1.2 Problem Statement, BR-x prefixes, Indirect Stakeholders Matrix, refined FRs (08, 14, 24, 25), updated IEEE citations, expanded Section 6.2 legal compliance, and adjusted cross-references.	1.1

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the complete set of functional and nonfunctional requirements for the TripEasy Tourism Management System, Version 1.1 this document has been prepared in accordance with IEEE Std 830-1998 **[1]**. A web-based application designed to assist tourists in planning one-day visits to attractions located within a 25 km radius of Ragama, Gampaha District, Sri Lanka. Data source baselines follow the Sri Lanka Tourism Development Authority guidelines **[2]** and local directories **[3]**.

The system provides attraction information, route guidance, interactive location mapping, category-based filtering, and day-visit itinerary planning capabilities for tourist users, together with a secure administrator portal for the management of attraction data. This document describes the complete system; it is not limited to a single subsystem or module and shall serve as the authoritative reference for developers, testers, the project supervisor, and project evaluators throughout the software development lifecycle.

1.2 Problem Statement

Tourists visiting the Gampaha District—particularly those seeking single-day excursions from transport hubs like Ragama—face significant challenges due to fragmented, unverified, and decentralized tourism information. Currently, attraction details, operating hours, travel tips, and route mapping are scattered across inconsistent online blogs, social media pages, or informal local sources. Furthermore, lesser-known cultural, historical, and recreational sites within the region lack digital visibility compared to major national tourist hubs.

As a result, tourists struggle to estimate travel distances accurately, evaluate realistic travel times between multiple locations, and organize multiple local attractions into a cohesive, single-day itinerary. Without a dedicated, centralized solution for the Gampaha area, tourists waste time on redundant route planning or miss out on nearby attractions altogether. The TripEasy Tourism Management System directly addresses these issues by consolidating attraction data, offering automated distance and travel time calculations, and providing interactive day-visit route planning within a defined 25 km radius of Ragama.

1.3 Document Conventions

This document has been prepared in accordance with the structure recommended by IEEE Std 830-1998. The following typographical and terminological conventions are used throughout this SRS:

- SRS refers to Software Requirements Specification.
- The keyword SHALL indicates a mandatory requirement that must be implemented.
- The keyword SHOULD indicates a recommended, but not strictly mandatory, requirement.
- The keyword MAY indicates an optional requirement or capability.
- Functional requirements are uniquely identified using the prefix FR-x (e.g., FR-01).

- Non-functional requirements are uniquely identified using the prefix NFR-x (e.g., NFR01).
- Business rules are uniquely identified using the prefix BR-x (e.g., BR-01).
- Each system feature described in Section 4 is assigned a priority of High, Medium, or Low, reflecting its relative importance to the project's core objectives.

1.4 Intended Audience and Reading Suggestions

This SRS is intended for the following categories of readers:

- Developers - responsible for the design and implementation of the system described herein.
- Testers - responsible for verifying that the implemented system satisfies the requirements stated in this document.
- Project Supervisor - responsible for evaluating the technical soundness and academic merit of the project.
- Stakeholders - including prospective tourism operators and end users with an interest in the system's capabilities.

Readers are advised to proceed through this document in the following sequence: (1) Introduction, to understand the purpose and scope of the project; (2) Overall Description, to gain a high-level understanding of the product and its context; (3) External Interface Requirements, to review the interfaces exposed by the system; (4) System Features, for the detailed functional requirements organized by feature; and (5) Other Nonfunctional Requirements, for the quality attributes and constraints that apply across the system.

1.5 Product Scope

TripEasy is a tourism information management system that centralizes information about tourist attractions located within the Gampaha District, with particular emphasis on destinations reachable as part of a single-day excursion from Ragama. The system is delivered as a responsive web application accessible through standard desktop and mobile browsers.

To address the difficulties outlined in the Problem Statement (Section 1.2), the key objectives of the TripEasy system are to:

- Promote local tourism within the Gampaha District by improving the visibility of lesserknown attractions.
- Provide tourists with accurate, centralized, and up-to-date attraction information.
- Support efficient day-trip planning through itinerary creation and distance/time estimation.
- Provide clear route visualization using an interactive mapping component.
- Enable authorized personnel to maintain attraction data through a dedicated administrator portal.

This SRS covers the complete TripEasy system, comprising the tourist-facing web interface, the administrator portal, the backend REST API, and the underlying database. It does not cover the

deployment infrastructure, hosting configuration, or any third-party services beyond their functional integration with the system.

2. Overall Description

2.1 Product Perspective

TripEasy is a new, self-contained web application; it is not a follow-on version of, nor a replacement for, any existing system. The product consists of four principal components operating together as a single, internally cohesive system:

- Frontend Interface - a ReactJS single-page application used by tourist users to browse, search, and plan visits to attractions.
- Admin Portal - a protected ReactJS interface used by administrators to manage attraction records.
- Backend REST API - a Node.js and Express.js service that exposes endpoints consumed by both the frontend interface and the admin portal, and that enforces business rules and authentication.
- MongoDB Database - a MongoDB Atlas cloud-hosted document database that persists attraction, category, and administrator data.

Figure 1 illustrates the high-level architecture of the system and the communication paths between its principal components.

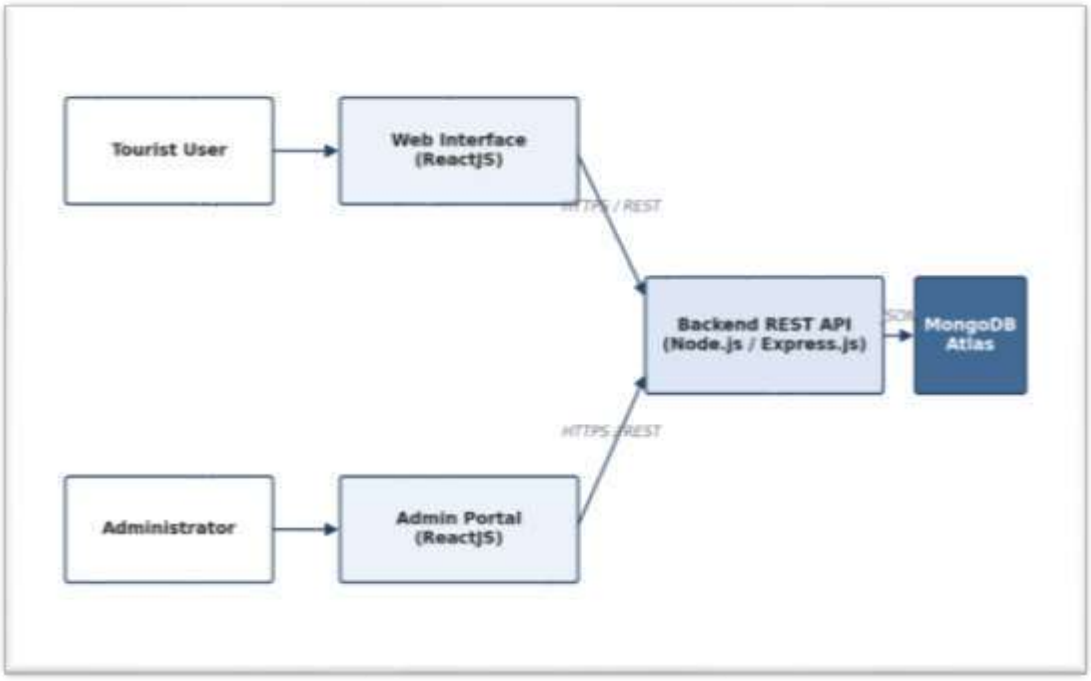


Figure 1**: TripEasy System Architecture**

2.2 Product Functions

The major functions provided by the TripEasy Tourism Management System are summarized below; detailed functional requirements for each are provided in Section 4.

- Attraction browsing and keyword search.
- Category-based filtering of attractions.
- Display of attraction locations on an interactive map.
- Viewing of detailed attraction information, including images, opening hours, and travel tips.
- Creation and management of a personalized day-visit itinerary.
- Route visualization and distance calculation between a reference point and selected attractions.
- Secure administrator login.
- Create, Read, Update, and Delete (CRUD) management of attraction records by administrators.

2.3 User Classes and Characteristics

2.3.1 Tourist User

Tourist users are members of the general public who access the system, typically without prior technical experience, to plan a day visit to attractions in the Gampaha District. This user class represents the primary and most frequent users of the system. Permitted actions include:

- Searching and browsing attractions.
- Viewing detailed attraction information.
- Creating and managing a personal day-visit itinerary.

2.3.2 Administrator

Administrators are authorized personnel responsible for maintaining the accuracy and completeness of attraction data within the system. This user class is smaller in number but requires elevated privileges. Permitted actions include:

- Logging in through a secure, authenticated session.
- Adding new attraction records.
- Updating existing attraction records.
- Deleting attraction records.
- Managing attraction categories.

2.3.3 Indirect Stakeholders

In addition to primary system users, the TripEasy system involves key indirect stakeholders who benefit from or influence the system's operations:

- **Attraction Owners and Local Authorities-** Local tourism operators, religious organizations, and local government bodies in the Gampaha District whose points of interest are featured on the platform. While they do not access the system directly, they benefit from increased tourist visibility and localized promotion.

- **Tourism Governance Bodies (SLTDA)-** Institutional bodies such as the Sri Lanka Tourism Development Authority (SLTDA) whose public directories and regulatory frameworks inform data accuracy and categorization.

Stakeholder Class	Stakeholder Type	Primary Responsibilities / System Interactions
Tourist User	Direct (Primary User)	Search/filter attractions, view details/maps, and assemble personalized day-visit itineraries.
Administrator	Direct (System Admin)	Manage system content via CRUD operations on attraction records and category definitions.
Developers & Testers	Internal Technical	System implementation, backend API development, map integration, and functionality verification.
Project Supervisor / Evaluators	Academic Evaluator	Review technical soundness, architecture, and academic merit of the system.
Attraction Owners / SLTDA	Indirect External	Provide baseline destination data; benefit from localized digital promotion.

Table 1:Stakeholder Summary Matrix

2.4 Operating Environment

Layer	Supported Environment
Client	Google Chrome, Mozilla Firefox, Microsoft Edge, and Apple Safari (latest two major versions), on desktop, tablet, and smartphone devices
Server	Node.js runtime with the Express.js web application framework
Database	MongoDB Atlas (cloud-hosted, managed NoSQL database service)

2.5 Design and Implementation Constraints

- The frontend shall be implemented using ReactJS.
- The backend shall be implemented using Node.js with the Express.js framework, exposing a RESTful API.
- Data shall be persisted in a MongoDB Atlas database.

- The user interface shall follow responsive design principles to support desktop, tablet, and smartphone form factors.
- An active internet connection is required for all system functions, as no offline mode is provided in Version 1.1.

2.6 User Documentation

The following user documentation components shall be delivered together with the software:

- User Manual - guidance for tourist users on browsing, searching, and itinerary planning.
- Administrator Guide - guidance for administrators on attraction and category management.
- API Documentation - reference material describing the backend REST endpoints.
- Installation Guide - instructions for deploying and configuring the system.

2.7 Assumptions and Dependencies

2.7.1 Assumptions

- Users have access to a device with a modern, JavaScript-enabled web browser.
- An active internet connection is available at the point of use.
- Tourism data supplied by the administrator remains substantially accurate and current.

2.7.2 Dependencies

- MongoDB Atlas, as the managed database service.
- React Leaflet, for rendering the interactive map component.
- A third-party hosting platform, for deployment of the frontend and backend services.

3. External Interface Requirements

3.1 User Interfaces

3.1.1 Tourist Dashboard

The tourist dashboard is the primary landing interface for tourist users and shall include:

- A search bar for locating attractions by name.
- Category filter controls.
- A scrollable grid or list of attraction cards, each displaying a thumbnail image, name, and category.
- An embedded interactive map.
- An itinerary panel summarizing attractions currently added to the user's day-visit plan.

3.1.2 Attraction Details Page

The attraction details page shall present comprehensive information for a single attraction, including:

- One or more images of the attraction.
- A descriptive summary of the attraction.
- Opening hours.
- Practical travel tips.
- Distance and estimated travel time from the reference point.

3.1.3 Admin Dashboard

The admin dashboard shall be accessible only to authenticated administrators and shall include:

- A secure login form.
- An attraction management table listing all attraction records.
- Create, update, and delete forms for attraction and category records.

3.2 Hardware Interfaces

The system does not interface directly with specialized hardware. It shall, however, operate correctly on the following classes of general-purpose computing devices, using the standard input mechanisms listed below.

Device Category	Input Mechanisms Supported
Desktop / Laptop computers	Keyboard, mouse
Tablet devices	Touchscreen
Smartphones	Touchscreen

3.3 Software Interfaces

The table below identifies the principal software components with which the TripEasy system interfaces, along with the purpose of each interface.

Table 1: Software Interfaces

Component	Purpose
ReactJS	Implements the tourist-facing and administrator user interfaces
Express.js	Implements the backend RESTful API
MongoDB Atlas	Provides persistent storage for attraction, category, and administrator data
React Leaflet	Renders the interactive map and attraction location markers
GitHub	Provides version control and source code management for the project

3.4 Communication Interfaces

The system relies on standard web communication protocols for all client-server interaction:

- HTTP and HTTPS - used as the transport protocols between client applications and the backend server; HTTPS shall be used for all production traffic (see NFR-07).
- REST APIs - the backend exposes a set of RESTful endpoints for retrieving and managing attraction, category, and itinerary data.
- JSON - all data exchanged between the client applications and the backend API is formatted as JSON.

4. System Features

This section organizes the functional requirements of the TripEasy system by feature, consistent with the major services the product provides to its user classes. Each feature is presented with its priority and associated functional requirements, uniquely identified using the FR-x convention.

4.1 Attraction Discovery and Search

Priority: High

This feature enables tourist users to locate attractions of interest through browsing, keyword search, and category filtering, and to view detailed information about a selected attraction.

FR-01: The system shall display all available attractions on the tourist dashboard, including a thumbnail image, name, and category for each attraction.

FR-02: The system shall allow the tourist user to search attractions by name using a search bar.

FR-03: The system shall allow the tourist user to filter displayed attractions by category (e.g., Historical, Religious, Natural, Recreational, Cultural).

FR-04: The system shall display detailed information for a selected attraction, including its description, opening hours, entrance fee (where applicable), and travel tips.

FR-05: The system shall display one or more images for each attraction.

4.2 Interactive Map Feature

Priority: High

This feature provides spatial context for attractions and supports route awareness through an embedded interactive map.

FR-06: The system shall display the geographic location of each attraction as a marker on an interactive map.

FR-07: The system shall calculate and display the travel distance from the reference point (Ragama) to each attraction.

FR-08: The system shall display route information between the reference point and a selected attraction as a visual overlay line (polyline) on the interactive map, accompanied by calculated distance and travel time.

FR-09: The system shall allow the tourist user to zoom and pan the interactive map.

FR-10: The system shall allow the tourist user to select a marker on the map to view a summary of the corresponding attraction.

4.3 Day-Visit Itinerary Planning

Priority: Medium

This feature allows tourist users to assemble a personalized day-visit plan from the available attractions and to review estimated distance and time for that plan.

FR-11: The system shall allow the tourist user to add attractions to a personalized day-visit itinerary.

FR-12: The system shall allow the tourist user to remove attractions from the itinerary.

FR-13: The system shall calculate and display the total estimated travel distance for all attractions currently in the itinerary.

FR-14: The system shall calculate and display the total estimated time required to visit all attractions in the itinerary, which shall represent the cumulative travel duration between all selected attractions plus a default estimated stay duration (e.g., 45 minutes) per attraction.

FR-15: The system shall allow the tourist user to view an ordered summary list of the itinerary.

4.4 Administrator Authentication

Priority: High

This feature restricts access to attraction management functions to authorized administrators only.

FR-16: The system shall provide a secure login form requiring a username and password for administrator access.

FR-17: The system shall validate administrator credentials and grant access to the admin dashboard only upon successful authentication.

FR-18: The system shall allow an authenticated administrator to log out, terminating the active session.

4.5 Attraction and Category Management

Priority: High

This feature enables administrators to maintain the attraction data presented to tourist users.

FR-19: The system shall allow an authenticated administrator to add a new attraction record, including name, category, description, images, location coordinates, and opening hours.

FR-20: The system shall allow an authenticated administrator to update an existing attraction record.

FR-21: The system shall allow an authenticated administrator to delete an attraction record.

FR-22: The system shall allow an authenticated administrator to add, update, or delete attraction categories.

FR-23: The system shall allow an authenticated administrator to view a searchable and filterable list of all attraction records for management purposes.

FR-24: The system shall display a clear "No attractions found" message and offer a filter reset button whenever a user search or category filter yields zero matching results.

FR-25: The system shall maintain itinerary items within the user's active browser session storage without requiring continuous user account login in Version 1.1.

Table 2: Functional Requirements Summary

Feature	Requirement IDs	Priority
Attraction Discovery and Search	FR-01 - FR-05 FR-24	High
Interactive Map Feature	FR-06 - FR-10	High
Day-Visit Itinerary Planning	FR-11 - FR-15 FR-25	Medium
Administrator Authentication	FR-16 - FR-18	High
Attraction and Category Management	FR-19 - FR-23	High

5. Other Nonfunctional Requirements

5.1 Performance Requirements

NFR-01: The system shall load the tourist dashboard within 3 seconds under normal broadband network conditions (2 Mbps or greater).

NFR-02: The system shall support at least 100 concurrent users without a noticeable degradation in response time.

NFR-03: Search and category-filter operations shall return results within 2 seconds of the request being submitted.

5.2 Safety Requirements

NFR-04: Distance and route information presented to the tourist user shall be derived from verified mapping data to avoid presenting misleading travel guidance.

NFR-05: The system shall present a clear disclaimer advising tourist users to independently verify attraction opening hours and travel conditions before departure.

5.3 Security Requirements

NFR-06: Administrator passwords shall be stored using an industry-standard one-way hashing algorithm (e.g., bcrypt); passwords shall never be stored or transmitted in plain text.

NFR-07: All communication between client applications and the backend API shall be encrypted using HTTPS.

NFR-08: The admin dashboard shall be accessible only within a valid authenticated session, and sessions shall expire automatically after a defined period of inactivity.

5.4 Software Quality Attributes

NFR-09: Usability: The user interface shall be designed so that a tourist user can reach any attraction's detailed information from the dashboard in no more than three interactions (clicks or taps).

NFR-10: Reliability / Availability: The system shall maintain a minimum of 99% uptime, excluding scheduled maintenance windows.

NFR-11: Maintainability: The system shall follow a modular architecture with clearly separated frontend, backend, and database layers to facilitate future maintenance and feature extension.

NFR-12: Portability: The system's user interface shall be responsive and shall render correctly across desktop, tablet, and smartphone screen sizes, and across all browsers listed in Section 2.4.

5.5 Business Rules

BR-01: Only authenticated administrators shall be permitted to create, update, or delete attraction and category records.

BR-02 : Attraction data displayed to tourist users shall be limited to attractions located within a 25 km radius of Ragama, Gampaha District.

Table 3: Non-Functional Requirements Summary

Category	Requirement IDs
Performance	NFR-01 - NFR-03
Safety	NFR-04 - NFR-05
Security	NFR-06 - NFR-08
Software Quality Attributes	NFR-09 - NFR-12
Business Rules	BR-01–BR-02

6. Other Requirements

6.1 Database Requirements

The system shall use MongoDB Atlas, a document-oriented NoSQL database, to persist attraction records, category definitions, and administrator credentials. Attraction records shall be modeled as JSON-like documents to accommodate variable attributes such as multiple images and travel tips without requiring schema migration.

6.2 Legal and Compliance Requirements

The system shall comply with the following data protection and intellectual property standards:

- **6.2.1 Data Protection & Privacy:** The system handles administrator credentials in accordance with industry-standard data protection guidelines, ensuring passwords are never stored in plain text (as specified in NFR-06). As Version 1.0 does not require tourist users to register or submit personally identifiable information (PII), no personal user profiling data is captured or retained.
- **6.2.2 Intellectual Property & Media Copyright:** All images, descriptions, and media assets used within the application for attraction discovery shall be sourced either from public domain archives, licensed under Creative Commons, or utilized with explicit baseline attribution to the original content creators or local tourism authorities.
- **6.2.3 Data Retention & Licensing:** Public attraction metadata managed within MongoDB Atlas shall be maintained solely for localized tourism promotion. Administrative access logs and session data shall be purged after 90 days to maintain optimal system performance and compliance.

6.3 Internationalization

Version 1.1 of the system shall be presented in English only. Support for Sinhala and Tamil language interfaces is identified as a candidate enhancement for future releases (see Appendix C).

7. References

[1] IEEE Computer Society, *IEEE Recommended Practice for Software Requirements Specifications*, IEEE Std 830-1998, Oct. 1998.

[2] Sri Lanka Tourism Development Authority (SLTDA), *Tourism Statistics and Resource Directory*, 2024. [Online]. Available: https://www.sltda.gov.lk

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Appendix A: Glossary

Table 4: Glossary of Terms

Term	Definition
SRS	Software Requirements Specification
FR	Functional Requirement
NFR	Non-Functional Requirement
CRUD	Create, Read, Update, and Delete - the four basic operations of persistent data management
REST API	Representational State Transfer Application Programming Interface; an architectural style for web services
JSON	JavaScript Object Notation; a lightweight data-interchange format
Itinerary	A personalized, ordered plan of attractions a tourist user intends to visit during a single day
Attraction	A tourist destination or point of interest recorded within the TripEasy system
Administrator	An authorized user responsible for maintaining attraction and category data
Tourist User	An end user of the system seeking information to plan a day visit
MongoDB Atlas	A managed, cloud-hosted MongoDB database service
React Leaflet	A React component library used to render interactive maps

Appendix B: Analysis Models

This appendix presents supplementary analysis models that illustrate the intended structure and usage of the TripEasy system.

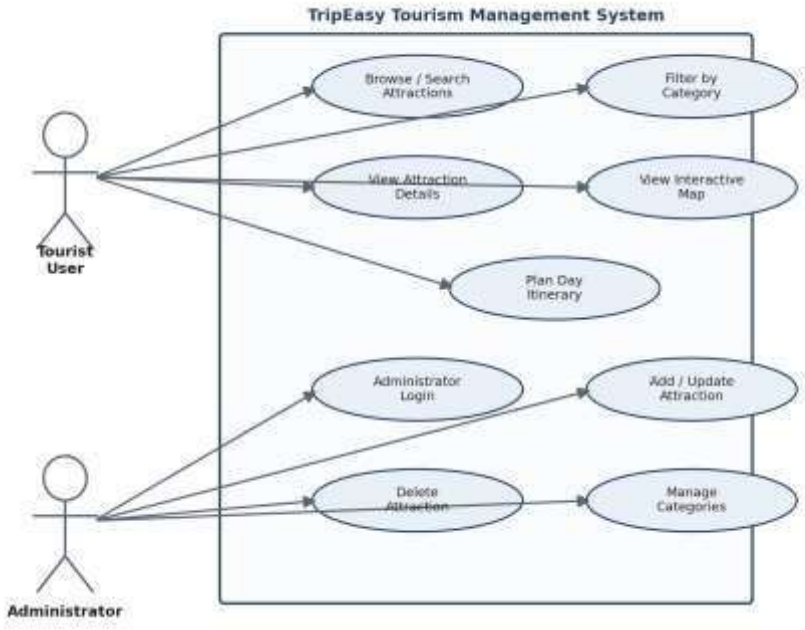


Figure 2**: Use Case Diagram - Tourist and Administrator Actors**

As shown in the use case diagram, the Tourist User actor interacts with the system to browse and search attractions, filter by category, view attraction details, view the interactive map, and plan a day itinerary. The Administrator actor interacts with the system to log in, add or update attractions, delete attractions, and manage categories. This model corresponds directly to the functional requirements presented in Section 4.

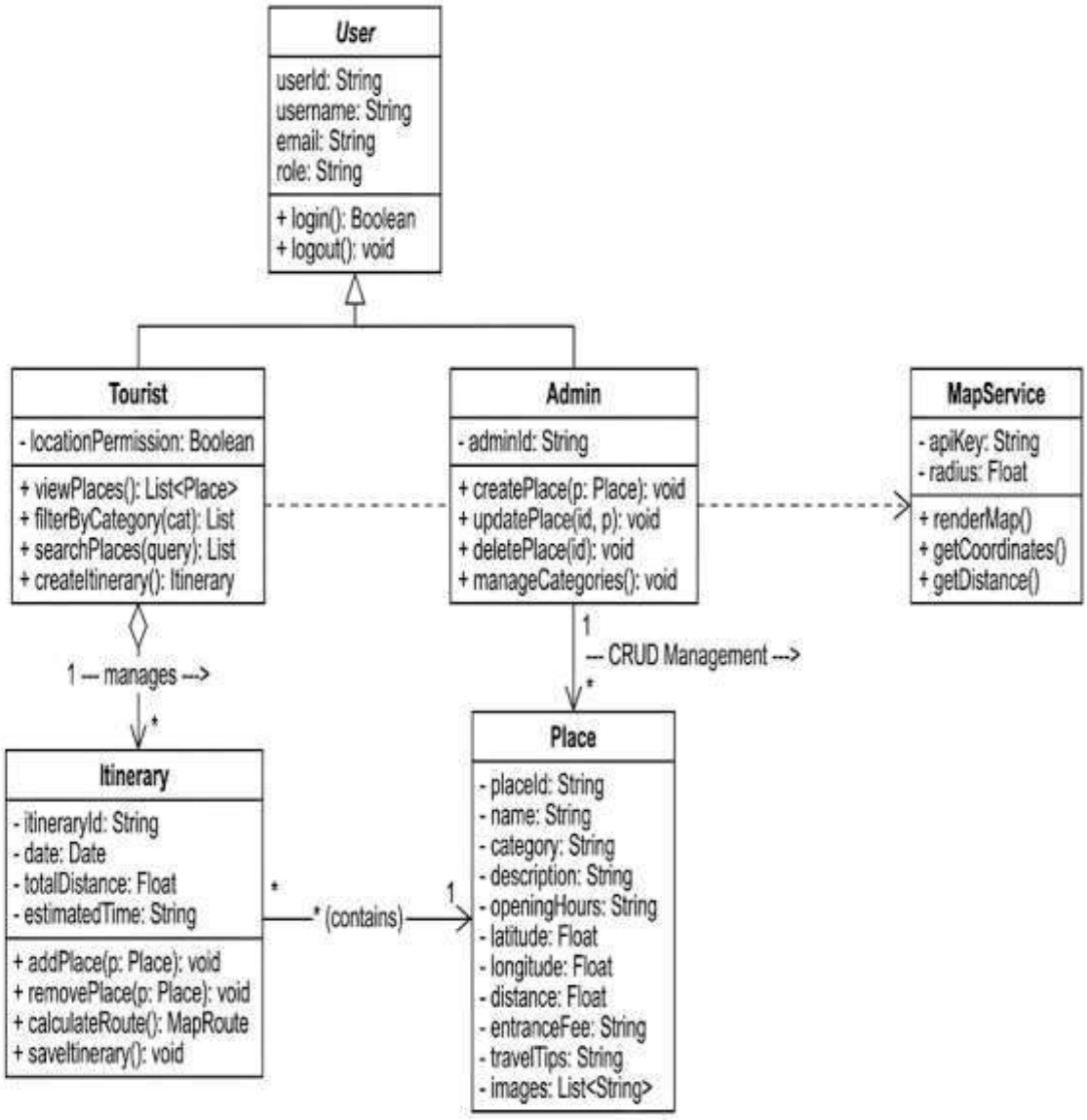


Figure 3 Class Diagram

Appendix C: To Be Determined List

The following items remain open at the time of this submission and are tracked here for closure in future iterations of the project:

#	TBD Item	Notes
TBD-1	Online payment integration for ticketed attractions	Out of scope for Version 1.0; candidate for future release
TBD-2	Multi-language support (Sinhala, Tamil)	See Section 6.3
TBD-3	Offline / low-connectivity mode	Requires further feasibility analysis
TBD-4	User accounts and saved itineraries for tourist users	Currently itineraries are session-based only